The Kalamazoo Zoo

It was a cold drizzly morning in early March 2010 when Rory Lyons, the director of the Kalamazoo Zoo, received the zoo's financial information for the year ending December 31, 2009.

The situation was quite different from what Rory had expected. As he folded his umbrella, he wondered whether the strange weather patterns were responsible for the big variances in the budget.

It had indeed been a peculiar year, weather-wise, in Kalamazoo. The winter had been the wettest on record, while the spring and summer had been unusually warm and balmy. The winter storms had forced the zoo to close for more days than originally planned. But the frisky spring weather had produced an unexpected success in the animal-breeding program.

Kalamazoo, a city in the state of Michigan, was facing a tough economic environment. This year, the zoo itself had received more state aid than expected, largely due to a renovation grant for its facilities. However, the overall budget situation was difficult. The state was reducing the amount of local aid it provided to the cities, forcing the city of Kalamazoo to reconsider many of its popular recreation activities. The Mayor had just called an emergency meeting to discuss the city's financial situation. Rory was asked to give a detailed presentation on the zoo's financial health and to present a preliminary budget for the next financial year.

This case was written by Linda Bilmes, Daniel Patrick Moynihan Senior Lecturer in Public Policy at the Harvard Kennedy School of Government. HKS cases are developed solely as the basis for class discussion. HKS Cases are not intended to serve as endorsements, sources of primary data, or illustrations of effective or ineffective management.

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Table 1: All Figures in $

Kalamazoo Zoo Financial Statement 2009

Revenues:	**ACTUAL**	**BUDGETED**
Gate Ticket Revenue from visitors	100,000	120,000
License Revenue from the food court	100,000	100,000
Donations from individuals	50,000	100,000
Grants for the Tiger Conservation Project*	180,000	150,000
Grants for the Rhino Conservation Project*	100,000	100,000
Grants for the Yellow-billed Cuckoo Conservation Project*	120,000	100,000
Subsidies from the state government	200,000	150,000
Total Revenues	850,000	820,000

* Funds received directly from Federal government

Expenditures:	**ACTUAL**	**BUDGETED**
Salary of Zoo Director	80,000	80,000
Salaries for Assistant Zoo Keepers (2 total)	100,000	100,000
Wages and Salary for Animal Handlers	100,000	100,000
Security, Office and Support staff Wages	50,000	50,000
Fringe Benefits cost for employees (health insurance, etc.)	130,000	130,000
Food and Provision costs for animals	360,000	240,000
Overtime costs	100,000	40,000
Utilities	50,000	30,000
Transportation and facilities for visitors	100,000	50,000
Total Expenditures	1,070,000	820,000

Question 1:

Perform a revenue and expenditure variance on the 2009 Kalamazoo budget based on the information provided in Table 1. State whether the revenue variance is favorable or unfavorable. State whether the expenditure variance is favorable or unfavorable. Use template #1 below to guide your analysis.

Template 1

Variance Type	**Actual**	**Budget**	**Variance Amount**	**Favorable/ Unfavorable**
Revenue Variance				
Expenditure Variance				

Table 2 : Operating Data

Item	Actual	Budgeted
Number of Visitors	10,000	15,000
Price per Admission Ticket	$10.00	$8.00
Number of donations	1000	500
Value of each donation	$50	$200
Number of animals in the Zoo	120	100
Food Consumed /animal/year	$3000	$2400
Overtime costs	$434/day	$120/day
Number of Zoo open days	230	250

Question 2:

Using the additional operating data on Kalamazoo provided in Table 2, compute revenue quantity variance and price variance for annual ticket revenue and state whether each is favorable or unfavorable. What do we know about the zoo from doing this analysis?

Template 2: Ticket Revenue Variance

Revenue Variance	Actual Quantity	Budgeted Quantity	Budgeted Price	Variance	Favorable/ Unfavorable
QUANTITY					

Revenue Variance	Actual Price	Budgeted Price	Actual Quantity	Variance	Favorable/ Unfavorable
PRICE					

Question 3:

Using the data provided in Table 2, compute expenditure quantity and price variance for animal food expenditures and state whether each is favorable or unfavorable. What do we learn about the Zoo from doing this analysis?

Template 3: Animal Food Expenditure Variance

Expenditure Variance	Actual Quantity	Budgeted Quantity	Budgeted Price	Variance	Favorable/ Unfavorable
Quantity					

Expenditure Variance	Actual Price	Budgeted Price	Actual Quantity	Variance	Favorable/ Unfavorable
Price					

Question 4:

What is the overall situation at the zoo that we see from performing these variances? Reviewing these issues and the budget, what else should Rory Lyons investigate? What information is needed to be able to perform these analyses?

Question 5:

As he was working on his budget analysis, Rory Lyons received a phone call from the Mayor. The Mayor said that due to state budget cutbacks. The Zoo should expect to receive $100,000 less in state subsidies for the coming year. Faced with this situation, which of the following options would you recommend to Rory Lyons for closing the budget shortfall? Calculate the net budgetary impact for each of these options using the actual budget:

1) Increase ticket prices to $15.00.

Rory believes this might reduce the number of visitors to the Zoo by 20%.

2) Reduce the number of animals to 100 by finding other host programs. There would be a one-time cost of transportation of $1000 each.

3) Fire one of the two assistant zookeepers. This will require a severance payment equal to 10% of the annual salary and a payment of full fringe benefits for 6 months. Note: the zookeepers are *not* the same people as the animal handlers.